A partition of the rotation group into two sets each of which squares onto the whole group

The Open Logic Project contributors

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Editorial scope. Single lemma card cut from the Open Logic Project unit olp-set-theory-choice (pinned-original-source.tex); statement and proof are the author's own text at source lines 961-965 and 967-974.

Lemma 0.1. *There is a partition of* $\mathrm{SO}(2)$ *into two disjoint sets,* $\mathrm{SO}(2)_1$ *and* $\mathrm{SO}(2)_2$*, both of which are a basis for* $\mathrm{SO}(2)$*.*

Proof. Let $\mathrm{SO}(2)_1$ consist of the rotations by rational radian values in $[0, \pi)$; let $\mathrm{SO}(2)_2 = \mathrm{SO}(2) \setminus \mathrm{SO}(2)_1$. By elementary algebra, $\{ \circ\rho\rho \mid \rho \in \mathrm{SO}(2)_1 \} = \mathrm{SO}(2)$. A similar result can be obtained for $\mathrm{SO}(2)_2$. □